

EX. NO: 7 — IMPLEMENTATION OF RSA

Aim:

To implement the RSA encryption algorithm using C.

Algorithm:

1. Choose two prime numbers p and q .
2. Calculate $n = p \times q$.
3. Calculate $\phi(n) = (p-1)(q-1)$.
4. Choose a value e relatively prime to $\phi(n)$.
5. Calculate private key d .
6. Encrypt using $C = M^e \bmod n$.
7. Decrypt using $M = C^d \bmod n$.

Program:

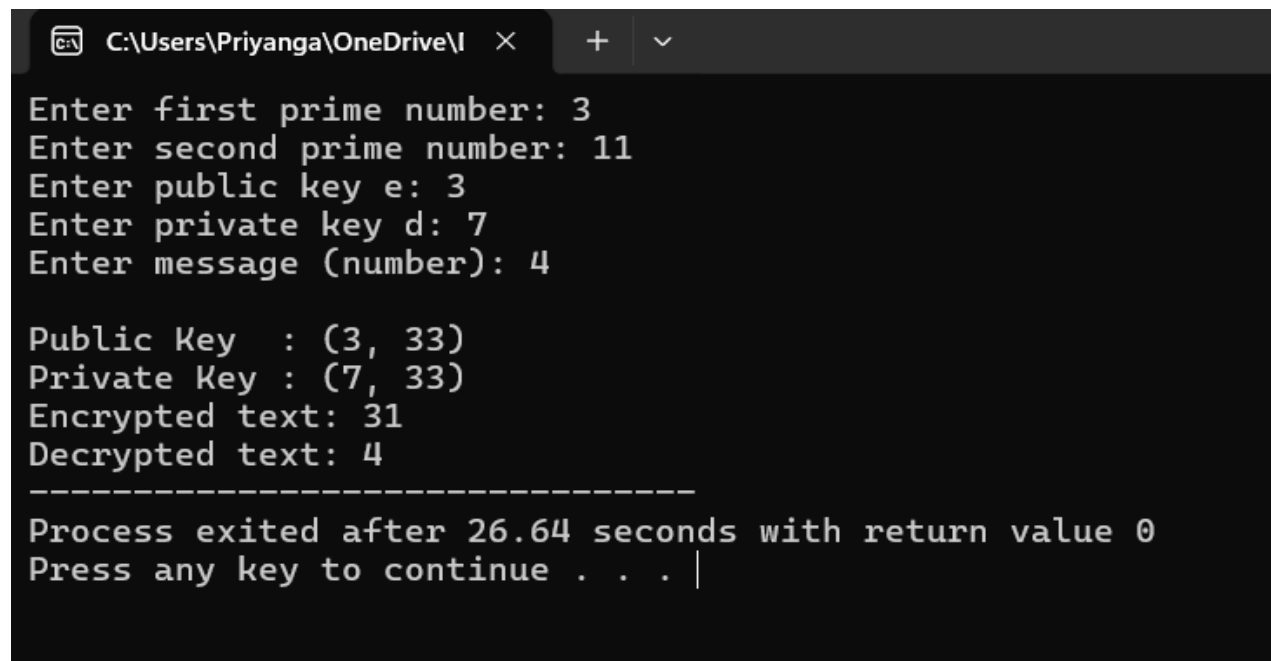
```
#include <stdio.h>

long long powerMod(long long base, long long exp, long long mod)
{
    long long result = 1;
    while(exp > 0)
    {
        result = (result * base) % mod;
        exp--;
    }
    return result;
}

int main()
{
```

```
long long p, q, n, phi, e, d;
long long message, encrypted, decrypted;
printf("Enter first prime number: ");
scanf("%lld", &p);
printf("Enter second prime number: ");
scanf("%lld", &q);
n = p * q;
phi = (p - 1) * (q - 1);
printf("Enter public key e: ");
scanf("%lld", &e);
printf("Enter private key d: ");
scanf("%lld", &d);
printf("Enter message (number): ");
scanf("%lld", &message);
encrypted = powerMod(message, e, n);
decrypted = powerMod(encrypted, d, n);
printf("\nPublic Key : (%lld, %lld)", e, n);
printf("\nPrivate Key : (%lld, %lld)", d, n);
printf("\nEncrypted text: %lld", encrypted);
printf("\nDecrypted text: %lld", decrypted);
return 0;
}
```

Output:

A screenshot of a Windows command prompt window with a dark background. The title bar shows the file path 'C:\Users\Priyanga\OneDrive\I' and standard window controls. The console text shows the execution of an RSA encryption program. It prompts for two prime numbers (3 and 11), a public key (3), and a private key (7). It then prompts for a message (4). The output shows the calculated public key (3, 33) and private key (7, 33), the encrypted text (31), and the decrypted text (4). A separator line of dashes follows. The program then reports it exited after 26.64 seconds and prompts for a key to continue.

```
C:\Users\Priyanga\OneDrive\I > Enter first prime number: 3
Enter second prime number: 11
Enter public key e: 3
Enter private key d: 7
Enter message (number): 4

Public Key : (3, 33)
Private Key : (7, 33)
Encrypted text: 31
Decrypted text: 4
-----
Process exited after 26.64 seconds with return value 0
Press any key to continue . . . |
```

Result:

Thus, the RSA encryption algorithm was implemented successfully using C.